

Compare GANs and Diffusion Models

Generative Adversarial Networks (GANs):

Introduced by Ian Goodfellow in 2014, GANs consist of two neural networks: a Generator and a Discriminator. The generator creates synthetic data, while the discriminator evaluates its authenticity. This adversarial process continues until the generator produces data indistinguishable from real data.

Diffusion Models:

Diffusion Models are a class of generative models that learn to reverse a gradual noising process to generate data. Starting from pure noise, these models iteratively denoise to produce coherent outputs. They have gained prominence for their ability to generate high-quality images.

Diffusion models start with random noise and slowly "clean" it step-by-step to create a realistic image. They learn how to reverse a process that adds noise to images, so they can generate new ones from pure noise.

Platform used to get GAN Model image:

<https://thispersondoesnotexist.com/>

Image:



Platform used to get Diffusion Model image:

<https://openart.ai/create>

Image:



Comparison of GAN vs Diffusion Models in Face Generation

1. Image Quality

GAN (StyleGAN2):

- Very high realism, resembling real photos
- Crisp edges with natural skin texture and lighting
- Excellent facial symmetry
- Minimal artifacts, though slight inconsistencies may appear

Diffusion (DALL·E/Stable Diffusion):

- Highly realistic, with smooth and aesthetic textures
- Soft edges, slightly airbrushed
- Excellent symmetry
- Very clean output with minimal artifacts

Winner: Tie

2. Training Stability

GAN (StyleGAN2):

- Training can be unstable due to adversarial nature
- Prone to mode collapse
- Requires fine-tuning and longer convergence

Diffusion Models:

- Very stable training via noise removal
- Rarely suffers from mode collapse
- Easier and more predictable convergence

Winner: Diffusion Models

3. Diversity of Results

GAN (StyleGAN2):

- Can produce varied outputs but requires latent space exploration
- Limited prompt control
- More focused on faces

Diffusion Models:

- Highly diverse and flexible
- Prompt-based generation allows custom outputs
- Can generate a wide range beyond faces

Winner: Diffusion Models

4. Ease of Use

GAN (StyleGAN2):

- Simple to use (just refresh page)
- No customization options
- Beginner-friendly interface

Diffusion Models:

- Requires prompt input
- Highly customizable
- Also beginner-friendly with creative freedom

Winner: Diffusion Models

Conclusion Table

Parameter	Winner
Image Quality	Tie
Training Stability	Diffusion Models
Diversity	Diffusion Models
Ease of Use	Diffusion Models